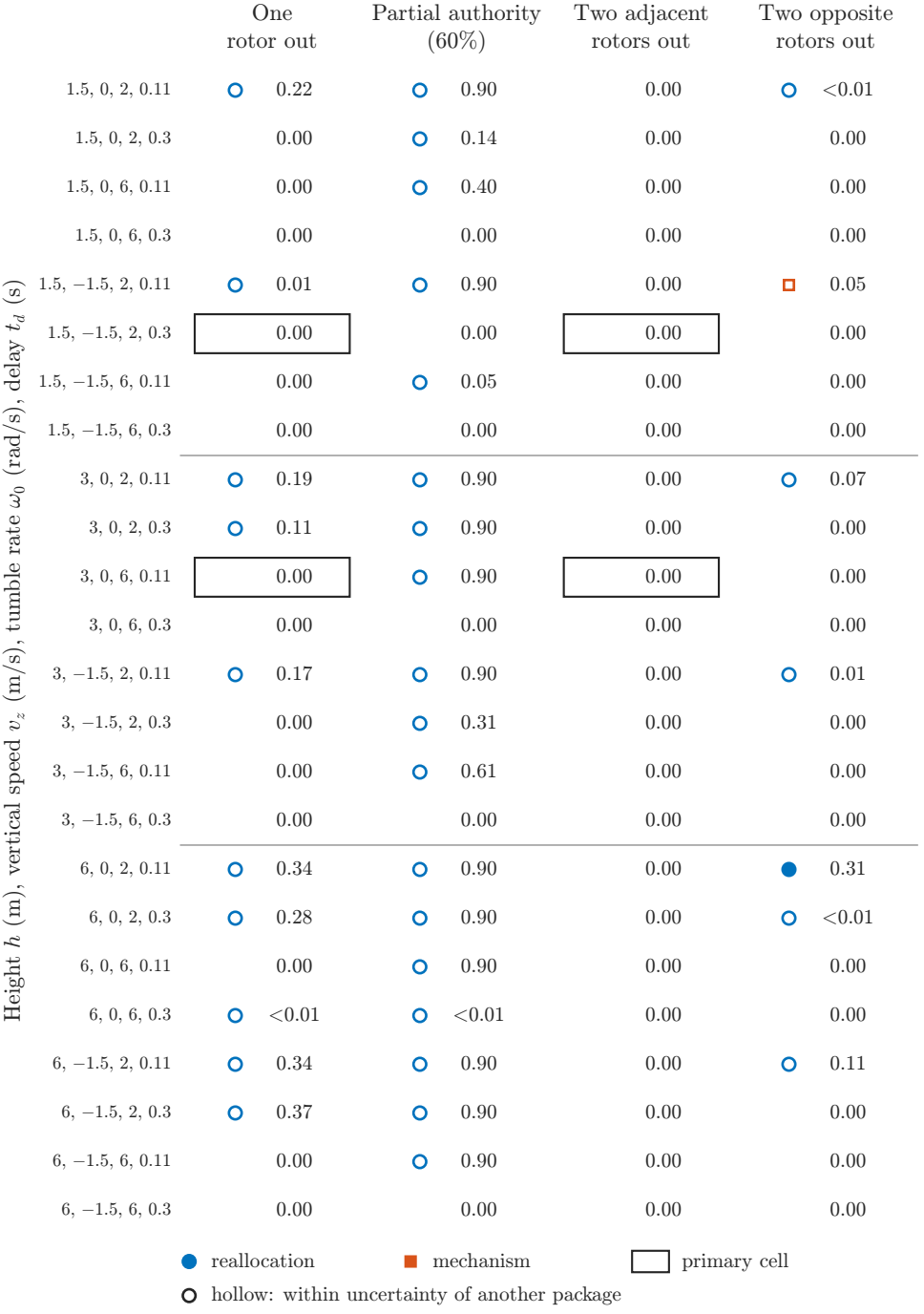
One rotor out
Partial authority (60%)
Two adjacent rotors out
Two opposite rotors out
Height h (m), vertical speed v_z (m/s), tumble rate ω_0 (rad/s), delay t_d (s)
1.5, 0, 2, 0.11 0.22 0.90 0.00 <0.01
1.5, 0, 2, 0.3 0.00 0.14 0.00 0.00
1.5, 0, 6, 0.11 0.00 0.40 0.00 0.00
1.5, 0, 6, 0.3 0.00 0.00 0.00 0.00
1.5, −1.5, 2, 0.11 0.01 0.90 0.00 0.05
1.5, −1.5, 2, 0.3 0.00 0.00 0.00 0.00
1.5, −1.5, 6, 0.11 0.00 0.05 0.00 0.00
1.5, −1.5, 6, 0.3 0.00 0.00 0.00 0.00
3, 0, 2, 0.11 0.19 0.90 0.00 0.07
3, 0, 2, 0.3 0.11 0.90 0.00 0.00
3, 0, 6, 0.11 0.00 0.90 0.00 0.00
3, 0, 6, 0.3 0.00 0.00 0.00 0.00
3, −1.5, 2, 0.11 0.17 0.90 0.00 0.01
3, −1.5, 2, 0.3 0.00 0.31 0.00 0.00
3, −1.5, 6, 0.11 0.00 0.61 0.00 0.00
3, −1.5, 6, 0.3 0.00 0.00 0.00 0.00
6, 0, 2, 0.11 0.34 0.90 0.00 0.31
6, 0, 2, 0.3 0.28 0.90 0.00 <0.01
6, 0, 6, 0.11 0.00 0.90 0.00 0.00
6, 0, 6, 0.3 <0.01 <0.01 0.00 0.00
6, −1.5, 2, 0.11 0.34 0.90 0.00 0.11
6, −1.5, 2, 0.3 0.37 0.90 0.00 0.00
6, −1.5, 6, 0.11 0.00 0.90 0.00 0.00
6, −1.5, 6, 0.3 0.00 0.00 0.00 0.00
reallocation
mechanism
primary cell
hollow: within uncertainty of another package